

RIGOR-OMR: Relation-Aware Interpretable Geometry for Zero-Shot Cross-Domain Optical Music Recognition

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Abstract

Optical music recognition (OMR) systems achieve strong performance on in-domain benchmarks, yet whether such results translate into plug-and-play recognition on unseen score domains remains underexplored. We present RIGOR-OMR (Relation-aware Interpretable Geometry for Out-of-domain Recognition), a modular framework that leverages staff-normalized geometry to guide visual symbol detection and explicit relation reasoning, while reconstructing structured outputs without target-domain training or dataset-specific adaptation. To compare systems with heterogeneous outputs, we introduce a representation-agnostic pitch–duration event evaluation protocol. On Polish Scores, RIGOR-OMR achieves 17.083 Event-F1, representing relative improvements of 357.8% over SMT and 1182.4% over Zeus. On Debussy, RIGOR-OMR significantly outperforms SMT while performing comparably to Zeus. Ablation studies confirm that explicit relation reasoning is the primary source of these improvements. Extensive experiments across Polish Scores, Debussy, GrandStaff, and OLIMPiC establish RIGOR-OMR as an effective approach to zero-shot music-content recognition under substantial domain shift.

1 Introduction

Optical music recognition (OMR) converts score images into machine-readable musical notation. Unlike conventional optical character recognition, identifying the visual class of a mark is not sufficient to determine its musical meaning; that meaning also depends on its staff-relative position, attachment and grouping relations, voice assignment, and reading order (Calvo-Zaragoza, Jr., and Pacha 2021; Rebelo et al. 2012). Recent OMR systems increasingly formulate the task as image-to-sequence prediction, using Transformer-based decoders to generate compact symbolic encodings (Vaswani et al. 2017; Ríos-Vila, Calvo-Zaragoza, and Paquet 2024; Li et al. 2023). This formulation has extended recognition from monophonic incipits to piano systems and complete pages (Ríos-Vila, Calvo-Zaragoza, and Rizo 2023; Ríos-Vila et al. 2026). Nevertheless, most reported progress is achieved under in-domain conditions, where training and test data share similar engraving styles, page layouts, notation vocabularies, and

output conventions. A model trained on one renderer, score collection, vocabulary, or layout may therefore fail sharply when deployed on an unseen score domain.

This deployment gap arises from both visual and symbolic domain shift. Synthetic piano scores typically exhibit clean backgrounds, regular spacing, and constrained engraving, whereas scanned historical scores may contain uneven contrast, page texture, warped systems, typographic variation, and diverse annotations. The output space may shift at the same time: Humdrum-derived encodings, Linearized MusicXML (LMX), tuplets, voices, and staff markers differ in both token inventory and structural assumptions. In image-to-sequence systems, visual recognition, musical interpretation, and output serialization are tightly coupled within a domain-specific decoder. Since higher-level model representations tend to become increasingly task-specific (Pan and Yang 2010; Yosinski et al. 2014), the decoder and its notation vocabulary can become particularly brittle when either the visual domain or the target representation changes.

To address this problem, we introduce RIGOR-OMR, short for *Relation-aware Interpretable Geometry for Out-of-domain Recognition*, a modular framework for zero-shot recognition on previously unseen score domains. Rather than directly mapping a score image to a source-specific autoregressive vocabulary, RIGOR-OMR decomposes recognition into staff-normalized symbol detection, explicit relation reasoning, and deterministic event reconstruction. Staff geometry first establishes a normalized coordinate frame. DEIM-D-FINE then detects notation primitives, while optional SAM2-based masks refine local shape evidence. Explicit relations subsequently associate noteheads with stems, beams, and other structural elements, after which a deterministic backend reconstructs the recognized content and exports either MusicXML or an evaluation-compatible sequence.

Unlike end-to-end sequence models, RIGOR-OMR separates visual detection from musical interpretation and output serialization. Its visual detector can therefore be replaced without modifying the structured reasoning and reconstruction backend. Moreover, the intermediate representations make detection, pitch, duration, relation, ordering, and reconstruction errors independently observable. To compare RIGOR-OMR with systems that produce incompatible symbolic representations, we further introduce a representation-agnostic evaluation protocol that projects heterogeneous outputs into a

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common space of pitch–duration events. This protocol enables controlled comparison under matched inputs, with all target-domain decisions fixed before evaluation and without target-domain training or dataset-specific adaptation.

To sum up, our contributions are as follows:

- We propose RIGOR-OMR, a modular and interpretable framework for zero-shot OMR on previously unseen score domains. RIGOR-OMR combines a replaceable visual detection front end with a structured reasoning and deterministic reconstruction backend, without requiring target-domain training or dataset-specific adaptation.
- We introduce a staff-normalized geometric representation and an explicit relation reasoning mechanism that associates noteheads with stems, beams, and other notation primitives. Through inspectable intermediate representations, the proposed mechanism transforms local symbol detections into structured pitch–duration events. Full-factorial and edge-level ablations further identify explicit relation reasoning as the primary source of the observed improvements.
- We introduce a representation-agnostic pitch–duration event evaluation protocol that projects system outputs with heterogeneous vocabularies and serialization formats into a common musical-content space, enabling fair comparison under matched inputs and target-domain decisions fixed before evaluation.
- RIGOR-OMR achieves an Event-F1 of 17.083 on Polish Scores, corresponding to relative improvements of 357.8% over SMT and 1182.4% over Zeus. On Debussy, it significantly outperforms SMT while achieving performance comparable to Zeus.

2 Related Work

2.1 Datasets and Modular OMR

OMR research has historically decomposed a page into staff detection, symbol recognition, and notation reconstruction (Calvo-Zaragoza, Jr., and Pacha 2021; Rebelo et al. 2012). Dataset design reflects this decomposition. MUSCIMA++ provides handwritten symbols and explicit pairwise relations, enabling notation-graph experiments (Jr. and Pecina 2017). DeepScores and DeepScoresV2 provide dense synthetic annotations for tiny printed objects, instance segmentation, and detection (Tugener et al. 2018, 2021). Camera-PrIMuS supports end-to-end recognition under realistic image degradation but is primarily monophonic (Calvo-Zaragoza and Rizo 2018). RealScores and recent instance-segmentation studies further emphasize that real-world score images differ materially from synthetic renders (Shatri and Fazekas 2023, 2024).

Object-centric OMR benefits from detectors that preserve small, dense instances. DETR introduced set prediction with bipartite matching (Carion et al. 2020); RT-DETRv2, D-FINE, and DEIM improve multiscale sampling, localization, and convergence (Lv et al. 2024; Peng et al. 2025; Huang et al. 2025). Promptable segmentation offers a complementary route from boxes to boundaries. SAM established zero-shot promptable segmentation (Kirillov et al. 2023), while SAM2

extended the design and improved image efficiency (Ravi et al. 2025). RIGOR-OMR uses these models as reusable components, not as evidence that natural-image priors alone understand notation semantics. DINOv2 features support retrieval and crop-level auxiliary classification (Oquab et al. 2024), but all final claims are based on end-output metrics rather than embedding similarity.

2.2 End-to-End Pianoform Recognition

GrandStaff introduced a large pianoform corpus and a sequence-recognition formulation over kern-like encodings (Ríos-Vila, Calvo-Zaragoza, and Rizo 2023). Sheet Music Transformer extended image-to-sequence OMR beyond monophonic transcription (Ríos-Vila, Calvo-Zaragoza, and Paquet 2024), and full-page variants combined convolutional encoders, autoregressive decoders, synthetic curriculum learning, and target-domain fine-tuning (Ríos-Vila et al. 2026). TrOMR likewise targets polyphonic recognition with Transformer-based decoding (Li et al. 2023). OLIMPiC introduced LMX, aligned MusicXML-compatible output, scanned evaluation systems, and a tree-aware metric (Mayer et al. 2024). These systems demonstrate strong in-domain accuracy, but their results also motivate a controlled view of cross-domain behavior.

2.3 Transfer and Domain Shift

Transfer learning studies distinguish reusable lower-level representations from source-specific higher layers (Pan and Yang 2010; Yosinski et al. 2014). Domain-adversarial training seeks features that cannot discriminate source from target domains (Ganin et al. 2016), and domain-generalization work organizes interventions into data, representation, and learning strategies (Wang et al. 2021). RIGOR-OMR studies a narrower and practically common setting: an OMR system must operate on an unseen collection without target annotations, while appearance, layout, and output representation may shift simultaneously.

3 Methodology

Figure 1 illustrates the four-stage RIGOR-OMR pipeline: Staff Geometry Estimation, DEIM-D-FINE Symbol Detection and Staff-Aware Prompt Generation, SAM2 Mask Refinement, and Relation Parsing and Score Reconstruction. The framework first establishes a staff-normalized coordinate system, detects notation primitives, and converts each detection into geometry-conditioned segmentation prompts. Refined instance masks are then transformed into typed geometric relations for deterministic pitch–duration event assembly and structured score export. The complete class inventory, prompt templates, relation thresholds, and reconstruction rules are provided in the supplementary material.

3.1 Staff Geometry Estimation

Given a score image I , RIGOR-OMR detects horizontal staff-line candidates using projection and connected-component cues and groups approximately equally spaced lines into five-line staves. For staff k , the detected line coordinates $\{\ell_{k,j}\}_{j=1}^5$

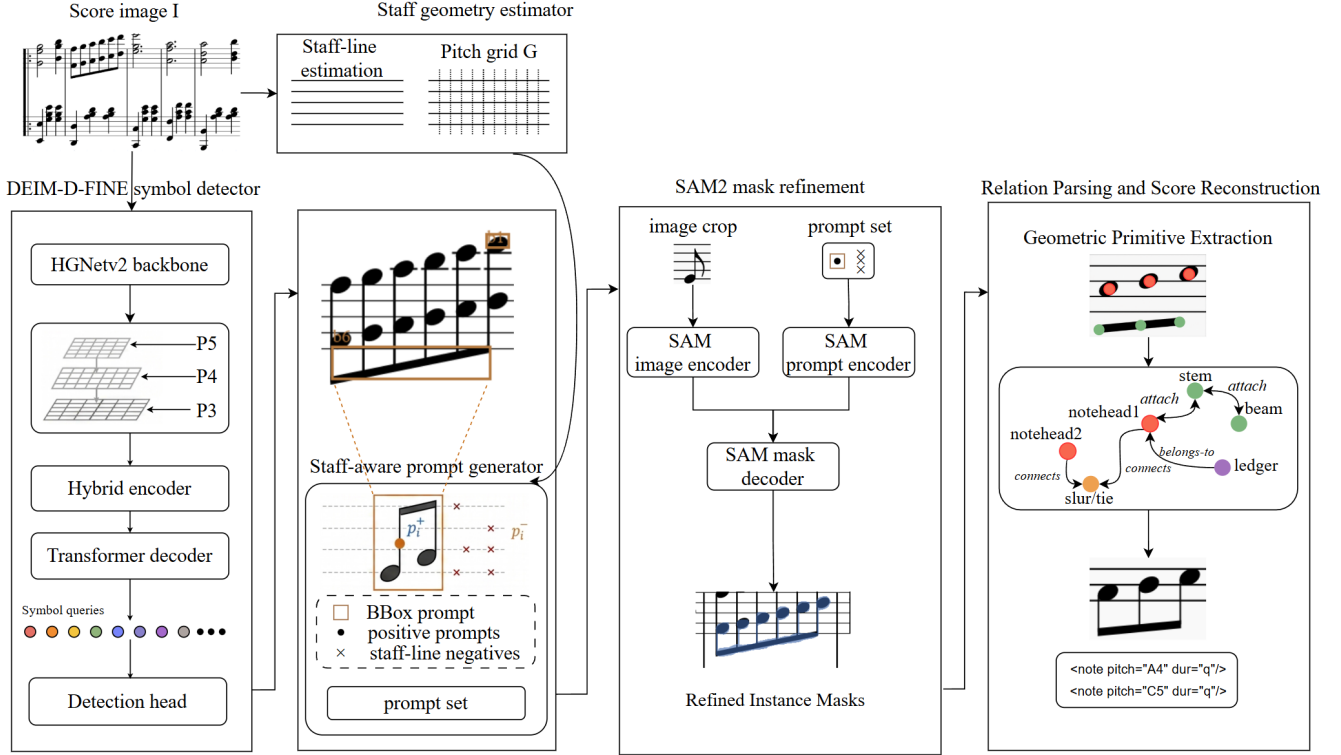


Figure 1: Overview of RIGOR-OMR. Staff geometry defines a normalized pitch grid; DEIM-D-FINE detects notation primitives and generates staff-aware prompts; SAM2 refines instance masks; and typed geometric relations support event assembly and structured score reconstruction.

define the staff spacing

$$s_k = \text{median}_{j=1}^4 (\ell_{k,j+1} - \ell_{k,j}). \quad (1)$$

All downstream distances, box sizes, and prompt offsets are normalized by s_k , reducing sensitivity to image resolution and engraving scale.

The five staff lines and their intermediate half-spacing positions form a discrete pitch grid G_k . Using the bottom staff line as reference, a notehead centered at vertical coordinate y_i is assigned the grid index

$$g_i = \text{round} \left(\frac{\ell_{k,5} - y_i}{0.5s_k} \right). \quad (2)$$

A deterministic clef-conditioned mapping

$$p_i = \Phi(c_k, g_i) \quad (3)$$

converts the grid index into an absolute pitch. The geometry stage outputs the staff regions, line coordinates, pitch grid, and normalized coordinate system used by prompt generation and relation parsing.

3.2 DEIM-D-FINE Symbol Detection and Staff-Aware Prompt Generation

The score image is processed by an OMR-specific DEIM-D-FINE detector consisting of an HGNetv2 backbone, a

multiscale hybrid encoder, and a Transformer decoder (Huang et al. 2025; Peng et al. 2025). Each retained detection is represented as

$$d_i = (b_i, c_i, q_i), \quad (4)$$

where b_i is the bounding box, c_i is the notation class, and q_i is the confidence score. Page-level and tiled predictions are mapped to the original image coordinates and merged using class-wise non-maximum suppression. Each symbol is then assigned to its nearest staff and expressed in the corresponding staff-normalized coordinate system.

For detection d_i , the staff-aware prompt generator extracts a local image crop I_i and constructs

$$\mathcal{P}_i = (\tilde{b}_i, \mathcal{P}_i^+, \mathcal{P}_i^-), \quad (5)$$

where $\tilde{b}_i$ is the box prompt in crop coordinates, while $\mathcal{P}_i^+$ and $\mathcal{P}_i^-$ are class-dependent positive and staff-line negative prompts.

Compact symbols such as noteheads use an interior foreground point. Elongated symbols such as stems and beams use multiple points along their principal direction. Negative points are sampled from staff lines intersecting the local region to suppress background line fragments, except for symbol families, such as noteheads, for which staff-line overlap is expected. All prompt locations are defined relative to the

detected box and s_k ; the exact templates are listed in the supplementary material.

3.3 SAM2 Mask Refinement

The local crop and its prompt set are passed to the SAM2 image encoder, prompt encoder, and mask decoder (Ravi et al. 2025):

$$\widehat{M}_i = \text{SAM2}(I_i, \mathcal{P}_i). \quad (6)$$

SAM2 refines the spatial support of each detected instance but does not modify its semantic class c_i . Consequently, visual classification remains the responsibility of DEIM-D-FINE, while SAM2 provides more accurate contour, orientation, skeleton, and endpoint evidence.

Area-like symbols are represented by their contours and support regions. For elongated symbols, the refined masks are skeletonized and summarized by a principal direction and two endpoints. If refinement fails, the corresponding detection box is used as a fallback support region. This design keeps mask refinement separable from the subsequent relation and reconstruction modules.

3.4 Relation Parsing and Score Reconstruction

Each symbol is represented as a typed node

$$v_i = (c_i, k_i, b_i, \widehat{M}_i, z_i, q_i) \in V, \quad (7)$$

where z_i denotes its class-specific geometric primitive. RIGOR-OMR then constructs a typed relation graph

$$\mathcal{G} = (V, E). \quad (8)$$

For relation type r , candidate edges are defined by

$$E_r = \{(v_i, v_j) : (c_i, c_j) \in \mathcal{C}_r, \phi_r(v_i, v_j; s_k) \leq \tau_r\}, \quad (9)$$

where $\mathcal{C}_r$ specifies compatible symbol classes and ϕ_r combines staff-normalized distance, overlap, endpoint, and orientation cues. The principal relation families connect noteheads to stems, stems to beams, ledger lines to noteheads, and slur/tie endpoints to noteheads.

For example, notehead–stem candidates require the stem endpoint or principal axis to lie sufficiently close to the notehead support. Stem–beam candidates require a stem endpoint to intersect or approach the beam support. Type-specific candidates are resolved using deterministic local priorities rather than a learned graph optimizer. Exact geometric conditions and conflict-resolution rules are reported in the supplementary material.

The retained relations bind independently detected attributes to the same musical event. For notehead i , pitch is obtained from Equation 2, while duration is determined by a deterministic mapping

$$D_i = \Psi(c_i, E_i^{\text{stem}}, E_i^{\text{beam}}, E_i^{\text{dot}}), \quad (10)$$

where Ψ combines notehead type, stem attachment, beam level, and augmentation-dot evidence. Accidentals and dots are assigned using staff-constrained nearest-neighbor rules.

Finally, the reconstruction backend groups events into staves, measures, onsets, and chords and serializes them in a deterministic score order. It exports both an auditable

pitch–duration event representation and structured outputs such as MusicXML or dataset-compatible proxy sequences. Since reconstruction is deterministic once V and E are fixed, individual relation families can be removed and the score rebuilt without retraining or rerunning the visual modules, enabling the controlled edge-level ablations in Section 4.6.

4 Experiments

4.1 Problem Formulation and Evaluation Protocol

Let $\mathcal{D}_s$ denote the source domain used to train a recognition component and $\mathcal{D}_t = \{(x_i^t, y_i^t)\}$ an unseen target domain. We study zero-shot transfer: neither model parameters nor target-specific thresholds are selected using samples from $\mathcal{D}_t$. Each x is a piano system or page and y is structured notation. The central difficulty is that methods emit heterogeneous sequences. We therefore project every valid output to a multiset of pitch–duration events and compute corpus-level precision, recall, and F1. If M is the multiset intersection count, P and G the predicted and reference event counts, then

$$\text{EventF1} = 100 \frac{2M}{P + G}. \quad (11)$$

This projection measures content recovery without rewarding a method for its native tokenization. It deliberately ignores event order, voice, and hierarchy; we retain SER and normalized tree-edit distance (TEDn) on OLiMPiC to measure those stricter properties.

The protocol has four safeguards. First, all methods receive the same target images and deterministic preprocessing. Second, no target test image is used for training, threshold selection, or checkpoint selection. Third, empty and failed outputs remain in the denominator. Fourth, all baseline values in the unified tables come from local inference and conversion, rather than copied headline numbers. We use 10,000 paired bootstrap resamples over pages or systems and report 95% intervals for F1 differences. A difference is called significant only when its interval excludes zero.

4.2 Setup and Baselines

Polish test24 and Debussy transfer24 are the main zero-shot targets. Debussy contains 24 evenly spaced systems from 24 distinct source documents. RIGOR-OMR, SMT (Ríos-Vila, Calvo-Zaragoza, and Paquet 2024), and Zeus (Mayer et al. 2024) receive identical target images and no target-domain adaptation. SMT uses its released FP-GrandStaff checkpoint and Zeus its released GrandStaff-LMX checkpoint. We additionally evaluate 200 frozen GrandStaff and 200 OLiMPiC outputs for relation ablation. All Event-F1 values are corpus aggregates; confidence intervals use 10,000 paired bootstrap resamples.

Table 1: Evaluation domains and their experimental roles.

Dataset	Target domain	N	Role
Polish	historical scans	24	main comparison
Debussy	unseen editions	24	comparison/ablation
GrandStaff	clean rendered	200	relation transfer
OLiMPiC	mixed systems	200	strict/relations

4.3 Zero-Shot Cross-Domain Content Recovery

Table 2 gives the central comparison. On Polish, RIGOR-OMR recovers 1,589 matching pitch–duration events versus 81 for Zeus and reaches 17.083 Event-F1, exceeding SMT by 13.352 points (95% CI [10.891, 15.774]) and Zeus by 15.751 points ([13.477, 18.422]). This is a measured zero-shot result under the same order-invariant protocol, not an oracle comparison. On Debussy it exceeds SMT by 6.449 points ([4.200, 8.585]). Its 0.736-point advantage over Zeus is not significant ([−2.465, 3.948]), so the supported conclusion is comparable performance rather than superiority.

Table 2: Zero-shot pitch–duration content recovery from local runs without target-domain adaptation.

Target	Method	Prec.	Rec.	F1	Pred./Gold
Polish	RIGOR-OMR	18.734	15.700	17.083	0.838
	SMT	2.733	5.879	3.731	2.151
	Zeus	3.971	0.800	1.332	0.202
Debussy	RIGOR-OMR	7.058	10.737	8.517	1.521
	SMT	1.277	5.432	2.068	4.252
	Zeus	7.463	8.126	7.781	1.089

The output ratios expose distinct failure modes. SMT generates more than four times the reference event count on Debussy, causing severe insertion error. Zeus emits only one fifth of the reference count on Polish, causing recall collapse. RIGOR-OMR remains imperfect but better balances recovered content and output size under the strongest shift. Figure 2 exposes both aggregate effects and uncertainty; Appendix Figure A1 gives the corresponding per-item distributions.

4.4 Strict Structure Is a Remaining Boundary

Order-invariant content recovery does not measure voice order or hierarchy. Table 3 therefore reports the official-style OLiMPiC test200 boundary. RIGOR-OMR completes every conversion and outperforms the failed SMT transfer, but Zeus remains substantially stronger in SER, Event-F1, and TEDn. This result bounds our claim to cross-domain content robustness rather than universal structured transcription superiority.

Table 3: Strict OLiMPiC test200 results. Lower SER/TEDn and higher Event-F1 are better.

Method	SER	No tup.	Event-F1	TEDn
RIGOR-OMR	74.421	73.574	8.786	61.342
SMT	100.000	–	0.000	100.000
Zeus	56.486	50.625	48.242	47.751

4.5 Full-Factorial Component Ablation

We evaluate all $2^4 = 16$ combinations of staff/geometry priors (A), symbol detection (B), optional SAM2 masks (C), and relation reasoning (D) on the 24 Debussy systems, totaling 384 predictions. Table 4 summarizes the informative configurations; Appendix Table 6 reports all 16. A and B are prerequisite stages. Removing D reduces F1 from 8.517 to 3.938, a 4.579-point loss with a paired 95% CI of [2.288, 7.019]. Removing C changes F1 by only +0.027 and its interval

[−0.614, 0.793] crosses zero. Thus relation reasoning, rather than segmentation masks, explains the principal measured gain.

Table 4: Debussy full-factorial ablation. A: geometry; B: detection; C: SAM2; D: relations.

Variant	Precision	Recall	Event-F1
A+B+C+D	7.058	10.737	8.517
A+B+D (without C)	7.113	10.695	8.544
A+B+C (without D)	8.437	2.568	3.938
A+B (without C,D)	8.437	2.568	3.938
Without A or B	0.000	0.000	0.000

4.6 Which Relations Transfer?

We next remove one edge family from frozen full-pipeline outputs and deterministically rebuild notation. Table 5 isolates the mechanism more precisely than switching the whole graph. Notehead–stem edges contribute significantly on all three datasets. Beam–stem edges are significant on Debussy and OLiMPiC; their positive GrandStaff effect is uncertain because the interval crosses zero. Slur/tie endpoints produce no Debussy Event-F1 change and are not presented as an effective component.

Table 5: Relation effects under frozen visual predictions. Δ is full minus ablated Event-F1; bold intervals exclude zero. Full scores appear in Appendix Table 9.

Target	Removed edge	Δ	95% CI
Debussy	notehead–stem	1.470	[0.282, 2.647]
Debussy	beam–stem	1.403	[0.212, 2.607]
Debussy	slur/tie	0.000	[0.000, 0.000]
GrandStaff	notehead–stem	0.669	[0.013, 1.321]
GrandStaff	beam–stem	0.583	[−0.067, 1.246]
OLiMPiC	notehead–stem	1.911	[1.472, 2.358]
OLiMPiC	beam–stem	1.543	[1.132, 1.972]

These ablations use the same cached A+B+C+D path within each dataset. Their OLiMPiC full value therefore belongs to the controlled ablation protocol and is not numerically pooled with the official-style value in Table 3.

4.7 Pitch–Duration Diagnosis

On Debussy, RIGOR-OMR obtains pitch-only and duration-only F1 values of 26.119 and 33.901, respectively, compared with 20.318 and 29.550 for Zeus and 4.553 and 16.530 for SMT. The much lower joint F1 of 8.517 shows that locally correct attributes are not reliably bound into the same event. RIGOR-OMR contains 527 pitch-correct/duration-wrong and 760 duration-correct/pitch-wrong matching opportunities. This localizes the next bottleneck in joint event binding and globally constrained decoding rather than in either marginal attribute alone. Figure 4 emphasizes the binding gaps; Table 7 retains the exact diagnostic values.

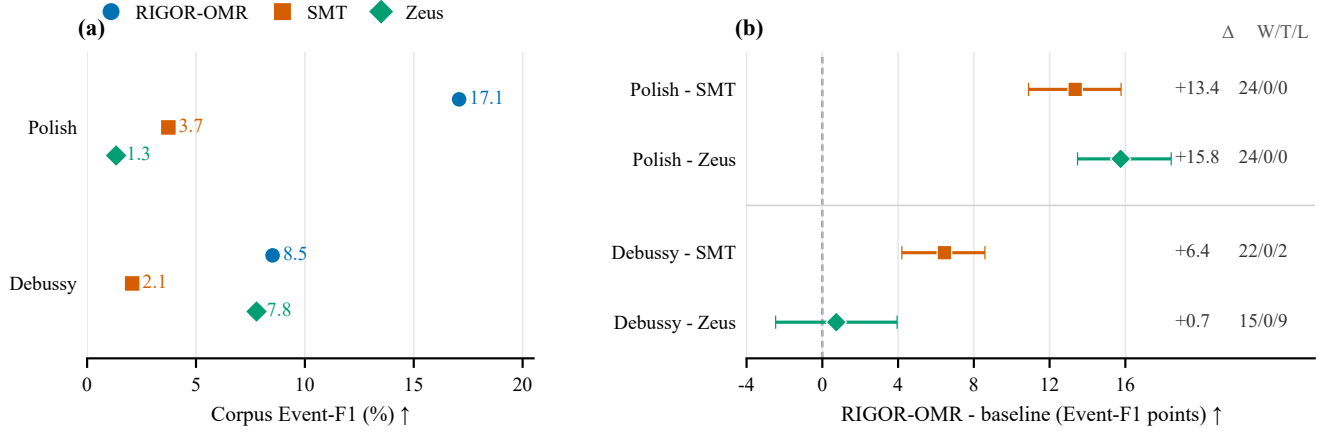


Figure 2: Zero-shot content recovery and paired uncertainty. (a) Corpus Event-F1 for the three frozen methods on Polish and Debussy. (b) RIGOR-OMR minus baseline differences with paired 95% bootstrap intervals and per-item win/tie/loss counts. The Polish gains and the Debussy gain over SMT exclude zero; the Debussy comparison with Zeus does not.

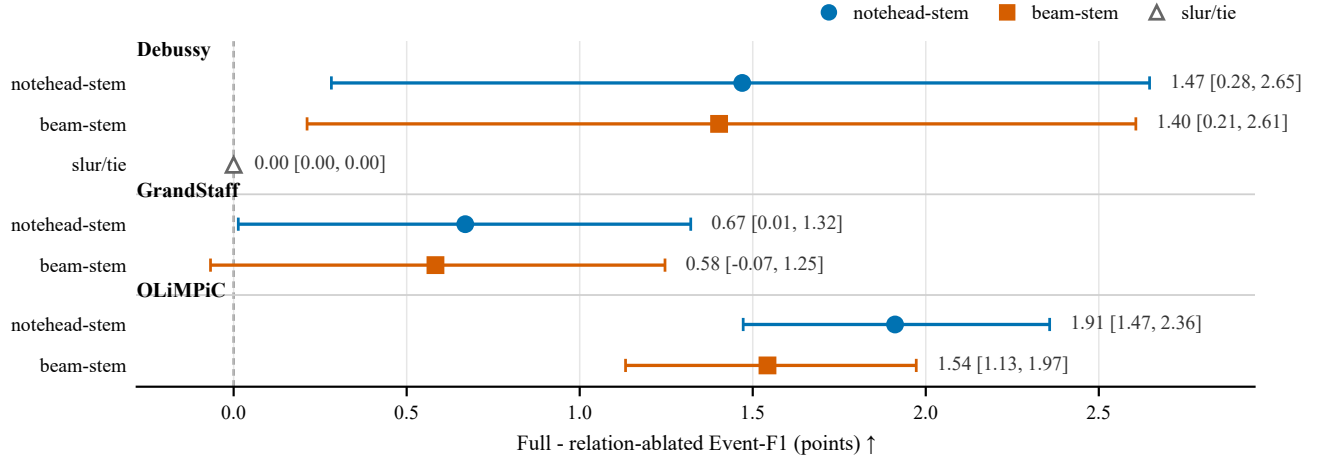


Figure 3: Cross-domain relation-family effects under frozen visual predictions. Points show full minus relation-ablated Event-F1 and bars show paired 95% bootstrap intervals. Notehead-stem edges transfer across all three datasets; beam-stem edges are reliable on Debussy and OLiMPiC but uncertain on GrandStaff; Debussy slur/tie endpoints are inactive under this metric.

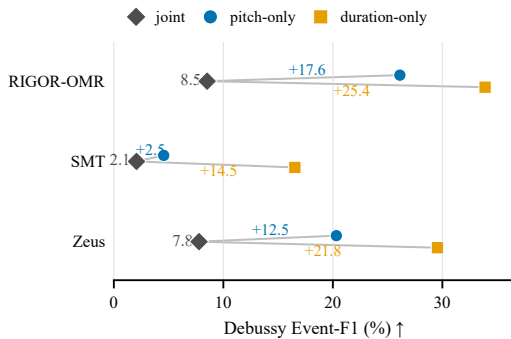


Figure 4: Attribute binding gaps on Debussy. Lines separately connect joint Event-F1 to pitch-only and duration-only recovery; labels give the corresponding marginal-minus-joint gaps. Marginal attribute recognition is substantially stronger than correct joint binding for every method.

4.8 Analysis

What the Results Establish The strongest positive evidence comes from severe zero-shot shift. On Polish, RIGOR-OMR recovers 1,589 matching events to Zeus’s 81 and improves Event-F1 from 1.332 to 17.083, with a paired interval that excludes zero; Debussy independently confirms a substantial advantage over SMT. The edge studies connect this behavior to a specific mechanism: staff-normalized notehead-stem reasoning is consistently useful across three targets, and beam-stem reasoning transfers on two. These effects arise without target-domain sequence training or visual reruns.

The evidence does not support a universal win over Zeus. Debussy is statistically tied, and OLiMPiC strongly favors Zeus under ordered and tree-structured metrics. The contrast reveals a useful division: explicit geometry preserves local musical content under appearance shift, whereas a learned autoregressive prior remains stronger at organizing events

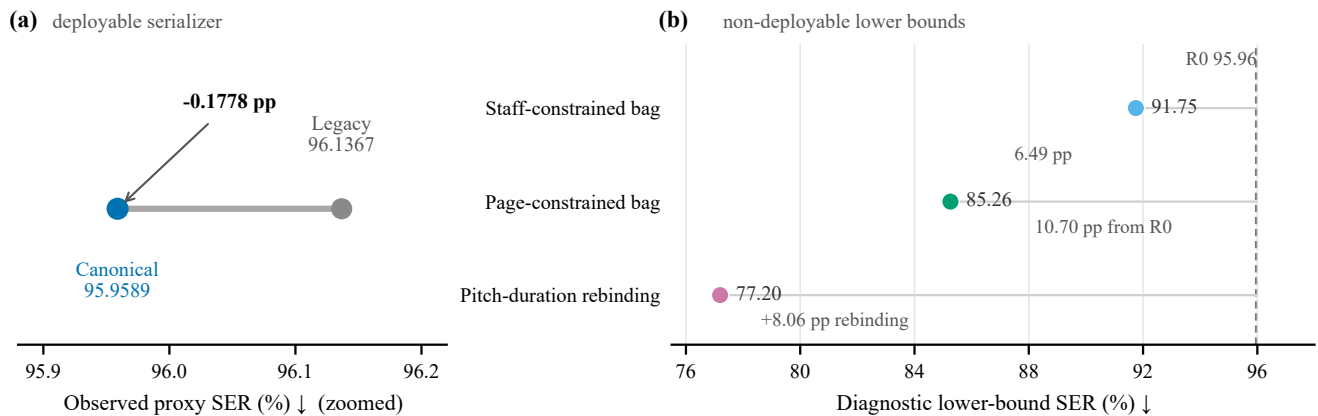


Figure 5: Structural diagnosis of the Polish eKern proxy evaluation. (a) Replacing the legacy staff-block serializer with a system-aware canonical order reduces SER by only 0.1778 points, showing that coarse reading order is not the dominant error source. (b) Page-constrained diagnostic lower bounds reveal substantially larger remaining spaces associated with staff assignment and pitch–duration binding. These lower bounds are diagnostic and are not deployable system results. The cross-page relaxed bag reference is excluded because it permits token matching across pages.

into globally valid voice and measure structure on a regular target.

Structural Error Is Not Just Serialization The Polish proxy evaluation provides a second view of the remaining structure gap. Replacing the legacy staff-block serializer with a canonical system-aware order lowers SER by only 0.1778 points. In contrast, page-constrained diagnostics expose much larger recoverable spaces when staff assignment, within-page matching, and pitch–duration binding are relaxed. These quantities are lower bounds under privileged matching, not deployable scores.

Inspectability and Failure Cases RIGOR-OMR retains staff lines, boxes, masks, typed edges, event hypotheses, and MusicXML, so failure analysis can identify which stage first loses information. Relation reasoning helps most systems but can harm complex pages when an incorrect edge propagates into duration and export. This behavior motivates confidence-weighted relation pruning and a globally constrained decoder, rather than adding more local edge types indiscriminately. Appendix Table 8 reports the complete structural diagnostic ledger, including low-coverage and cross-page references excluded from Figure 5.

4.9 Limitations and Responsible Use

Absolute zero-shot Event-F1 remains low, so the reported gains do not imply deployment-ready transcription. Event-F1 ignores order and hierarchy, and the OLiMPiC results show that these omissions are substantial. SAM2 has no verified end-metric benefit, slur/tie edges are currently inactive under the event projection, and detector replacement does not imply detector-insensitive performance. The 24-item Polish and Debussy sets also motivate broader evaluation. Finally, we have not completed a matched-hardware comparison of runtime, peak memory, or trainable parameters, so RIGOR-OMR’s modular adaptation cost remains a design property rather than an experimentally established efficiency advantage.

The datasets contain public-domain or research-licensed score images, but deployment on contemporary editions must respect copyright and dataset licenses. Inaccurate pitch or rhythm transcriptions should not be presented as authoritative; practical use should preserve the source image, expose confidence, and support human correction.

5 Conclusion

RIGOR-OMR studies OMR applicability under zero-shot domain shift rather than extrapolating from in-domain scores. Under a common pitch–duration event projection, it significantly outperforms SMT and Zeus on Polish and significantly outperforms SMT while matching Zeus on Debussy. Full-factorial and edge-level ablations attribute these gains to explicit relation reasoning, especially notehead–stem and beam–stem attachments. At the same time, OLiMPiC SER and TEDn show that local relational correctness has not yet become globally ordered, hierarchical notation. The resulting research direction is clear: preserve the transferable geometry and inspectability of RIGOR-OMR while adding constrained event binding and voice-aware structural decoding.

References

- Calvo-Zaragoza, J.; Jr., J. H.; and Pacha, A. 2021. Understanding Optical Music Recognition. *ACM Computing Surveys*, 53(4): 77:1–77:35.
- Calvo-Zaragoza, J.; and Rizo, D. 2018. Camera-PrIMuS: Neural End-to-End Optical Music Recognition on Realistic Monophonic Scores. In *Proceedings of the International Conference on Music Information Retrieval*, 248–255.
- Carion, N.; Massa, F.; Synnaeve, G.; Usunier, N.; Kirillov, A.; and Zagoruyko, S. 2020. End-to-End Object Detection with Transformers. In *European Conference on Computer Vision*, 213–229.
- Ganin, Y.; Ustinova, E.; Ajakan, H.; Germain, P.; Larochelle, H.; Laviolette, F.; Marchand, M.; and Lempitsky, V. 2016.

- Domain-Adversarial Training of Neural Networks. *Journal of Machine Learning Research*, 17(59): 1–35.
- Huang, S.; Lu, Z.; Cun, X.; Yu, Y.; Zhou, X.; and Shen, X. 2025. DEIM: DETR with Improved Matching for Fast Convergence. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Jr., J. H.; and Pecina, P. 2017. The MUSCIMA++ Dataset for Handwritten Optical Music Recognition. In *Proceedings of the International Conference on Document Analysis and Recognition*, 39–46.
- Kirillov, A.; Mintun, E.; Ravi, N.; Mao, H.; Rolland, C.; Gustafson, L.; Xiao, T.; Whitehead, S.; Berg, A. C.; Lo, W.-Y.; Dollár, P.; and Girshick, R. 2023. Segment Anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 4015–4026.
- Li, Y.; Liu, H.; Jin, Q.; Cai, M.; and Li, P. 2023. TrOMR: Transformer-Based Polyphonic Optical Music Recognition. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 1–5.
- Lv, W.; Zhao, Y.; Chang, Q.; Huang, K.; Wang, G.; and Liu, Y. 2024. RT-DETRv2: Improved Baseline with Bag-of-Freebies for Real-Time Detection Transformer. *arXiv preprint arXiv:2407.17140*.
- Mayer, J.; Straka, M.; Jr., J. H.; and Pecina, P. 2024. Practical End-to-End Optical Music Recognition for Pianoform Music. In *Proceedings of the International Conference on Document Analysis and Recognition*.
- Oquab, M.; Darcet, T.; Moutakanni, T.; Vo, H.; Szafraniec, M.; Khalidov, V.; Fernandez, P.; Haziza, D.; Massa, F.; El-Nouby, A.; Assran, M.; Ballas, N.; Galuba, W.; Howes, R.; Huang, P.-Y.; Li, S.-W.; Misra, I.; Rabbat, M.; Sharma, V.; Synnaeve, G.; Xu, H.; Jégou, H.; Mairal, J.; Labatut, P.; Joulin, A.; and Bojanowski, P. 2024. DINOv2: Learning Robust Visual Features without Supervision. *Transactions on Machine Learning Research*.
- Pan, S. J.; and Yang, Q. 2010. A Survey on Transfer Learning. *IEEE Transactions on Knowledge and Data Engineering*, 22(10): 1345–1359.
- Peng, Y.; Li, H.; Wu, P.; Zhang, Y.; Sun, X.; and Wu, F. 2025. D-FINE: Redefine Regression Task in DETRs as Fine-Grained Distribution Refinement. In *International Conference on Learning Representations*.
- Ravi, N.; Gabeur, V.; Hu, Y.-T.; Hu, R.; Ryali, C.; Ma, T.; Khedr, H.; Rädle, R.; Rolland, C.; Gustafson, L.; Mintun, E.; Pan, J.; Alwala, K. V.; Carion, N.; Wu, C.-Y.; Girshick, R.; Dollár, P.; and Feichtenhofer, C. 2025. SAM 2: Segment Anything in Images and Videos. In *International Conference on Learning Representations*.
- Rebelo, A.; Fujinaga, I.; Paszkiewicz, F.; Marçal, A. R. S.; Guedes, C.; and Cardoso, J. S. 2012. Optical Music Recognition: State-of-the-Art and Open Issues. *International Journal of Multimedia Information Retrieval*, 1: 173–190.
- Ríos-Vila, A.; Calvo-Zaragoza, J.; and Paquet, T. 2024. Sheet Music Transformer: End-to-End Optical Music Recognition Beyond Monophonic Transcription. In *Proceedings of the International Conference on Document Analysis and Recognition*, 20–37.
- Ríos-Vila, A.; Calvo-Zaragoza, J.; and Rizo, D. 2023. End-to-End Optical Music Recognition for Pianoform Sheet Music. *International Journal on Document Analysis and Recognition*, 26: 347–362.
- Ríos-Vila, A.; Calvo-Zaragoza, J.; Rizo, D.; and Paquet, T. 2026. End-to-End Full-Page Optical Music Recognition for Pianoform Sheet Music. *International Journal of Computer Vision*, 134(2).
- Shatri, E.; and Fazekas, G. 2023. Real World Music Object Recognition. *Transactions of the International Society for Music Information Retrieval*, 6(1).
- Shatri, E.; and Fazekas, G. 2024. Knowledge Discovery in Optical Music Recognition: Enhancing Information Retrieval with Instance Segmentation. *arXiv preprint arXiv:2408.15002*.
- Tuggener, L.; Elezi, I.; Schmidhuber, J.; Pelillo, M.; and Stadelmann, T. 2018. DeepScores: A Dataset for Segmentation, Detection and Classification of Tiny Objects. In *Proceedings of the International Conference on Pattern Recognition*, 3704–3709.
- Tuggener, L.; Satyawand, Y. P.; Pacha, A.; Schmidhuber, J.; and Stadelmann, T. 2021. The DeepScoresV2 Dataset and Benchmark for Music Object Detection. In *Proceedings of the International Conference on Pattern Recognition*, 9188–9195.
- Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, L.; and Polosukhin, I. 2017. Attention Is All You Need. In *Advances in Neural Information Processing Systems*, volume 30.
- Wang, J.; Lan, C.; Liu, C.; Ouyang, Y.; and Qin, T. 2021. Generalizing to Unseen Domains: A Survey on Domain Generalization. In *Proceedings of the International Joint Conference on Artificial Intelligence*, 4627–4635.
- Yosinski, J.; Clune, J.; Bengio, Y.; and Lipson, H. 2014. How Transferable Are Features in Deep Neural Networks? In *Advances in Neural Information Processing Systems*, volume 27.

A Complete Component Ablation

Table 6 reports every configuration in the Debussy 2⁴ factorial experiment. Configurations without geometry (A) or detection (B) produce no events. Among viable variants, relation reasoning (D) changes recall and output coverage substantially, whereas SAM2 mask refinement (C) has no measurable end-metric benefit under this protocol.

Table 6: Complete Debussy full-factorial ablation. Pred./Gold is the predicted-to-reference event-count ratio.

Variant	Prec.	Rec.	F1	Pred./Gold
A0B0C0D0	0.000	0.000	0.000	0.000
A0B0C0D1	0.000	0.000	0.000	0.000
A0B0C1D0	0.000	0.000	0.000	0.000
A0B0C1D1	0.000	0.000	0.000	0.000
A0B1C0D0	0.000	0.000	0.000	0.000
A0B1C0D1	0.000	0.000	0.000	0.000
A0B1C1D0	0.000	0.000	0.000	0.000
A0B1C1D1	0.000	0.000	0.000	0.000
A1B0C0D0	0.000	0.000	0.000	0.000
A1B0C0D1	0.000	0.000	0.000	0.000
A1B0C1D0	0.000	0.000	0.000	0.000
A1B0C1D1	0.000	0.000	0.000	0.000
A1B1C0D0	8.437	2.568	3.938	0.304
A1B1C0D1	7.113	10.695	8.544	1.504
A1B1C1D0	8.437	2.568	3.938	0.304
A1B1C1D1	7.058	10.737	8.517	1.521

B Additional Diagnostic Results

Figure A1 shows the paired per-page and per-system zero-shot comparisons. Figure A2 then stratifies Debussy systems by a predeclared reference-event density tertile. RIGOR-OMR is strongest on the middle-density group, while Zeus is strongest on the highest-density group; performance is therefore not a monotonic function of event count alone.

Table 7: Debussy oracle diagnostics using target annotations. These are diagnostic upper bounds, not deployable results.

Method	Joint	Pitch corr.	Dur. corr.	Count corr.
RIGOR-OMR	8.517	33.901	26.119	11.118
SMT	2.068	16.530	4.553	5.562
Zeus	7.781	29.550	20.318	9.234

C Polish Structural Diagnostic Ledger

Table 8 reports the complete proxy-SER diagnostic ladder. R3 is retained for auditability but has only 141 complete matches among 5002 gold onset groups. The cross-page relaxed bag is a mathematical reference only: because it permits tokens to match across page boundaries, it is not a valid page-constrained oracle.

D Relation Inventory and Reproducibility

All edge ablations reuse frozen A+B+C+D visual outputs, remove one relation family, and deterministically rebuild the structured prediction. Table 9 gives the complete scores, and Table 10 reports the number of instantiated edges and affected pages. Debussy confidence intervals use 10,000 paired

bootstrap resamples with seed 20260720; the GrandStaff and OLiMPiC reproduction gates complete all 200 examples without failed pages.

Table 9: Complete edge-level relation ablation under frozen visual predictions. Δ is full minus ablated Event-F1.

Target	Removed edge	Full	Ablated	Δ	95% CI
Deb.-24	note-stem	8.517	7.047	1.470	[0.282, 2.647]
Deb.-24	beam-stem	8.517	7.114	1.403	[0.212, 2.607]
Deb.-24	slur/tie	8.517	8.517	0.000	[0.000, 0.000]
GS-200	note-stem	8.887	8.217	0.669	[0.013, 1.321]
GS-200	beam-stem	8.887	8.304	0.583	[-0.067, 1.246]
OLi.-200	note-stem	9.624	7.713	1.911	[1.472, 2.358]
OLi.-200	beam-stem	9.624	8.082	1.543	[1.132, 1.972]

Table 10: Relation inventory in frozen full-pipeline outputs. Entries are edge count / pages with an edge.

Dataset	Note-stem	Beam-stem	Slur/tie	Ledger-note
Debussy-24	3900 / 24	585 / 23	128 / 21	inactive
GrandStaff-200	14947 / 200	2395 / 192	198 / 83	4 / 4
OLiMPiC-200	30113 / 200	333 / 111	266 / 117	8252 / 200

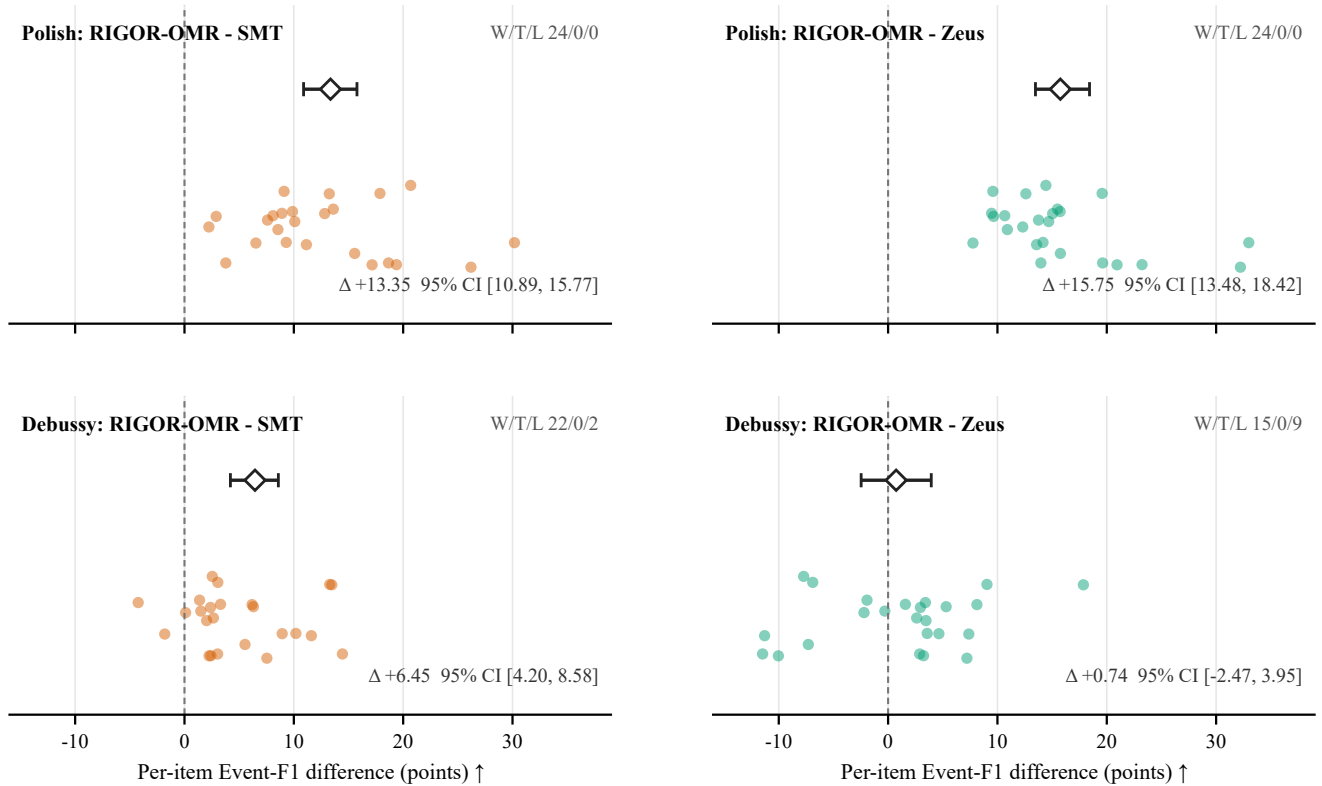


Figure A1: Per-item zero-shot Event-F1 differences. Positive values favor RIGOR-OMR. Dots are held-out pages or systems; diamonds and bars show the corpus difference and paired 95% bootstrap interval, with win/tie/loss counts reported in each panel. Polish separates cleanly from both baselines, while the Debussy comparison with Zeus overlaps zero.

Table 8: Complete Polish structural diagnostics. “Gain from R0” is $SER_{R0} - SER$, so positive values denote diagnostic headroom. Lower-bound rows use privileged matching and are not deployable system outputs.

Diagnostic	Constraint	SER↓	Gain from R0	Coverage	Page-valid	Interpretation
Legacy	staff-block order	96.1367	-0.1778	corpus	yes	observed serializer
R0 canonical	system-aware order	95.9589	0.0000	corpus	yes	observed, deployable
R2 staff bag	within staff	91.7498	4.2091	corpus	yes	diagnostic lower bound
R3 exact onset	exact onset groups	95.9293	0.0296	141/5002	yes	low-coverage diagnostic
Page bag	within page	85.2584	10.7005	corpus	yes	diagnostic lower bound
R6 rebinding	within-page pitch/duration	77.1959	18.7630	corpus	yes	diagnostic lower bound
Cross-page bag	across pages	57.0793	38.8796	corpus	no	mathematical reference

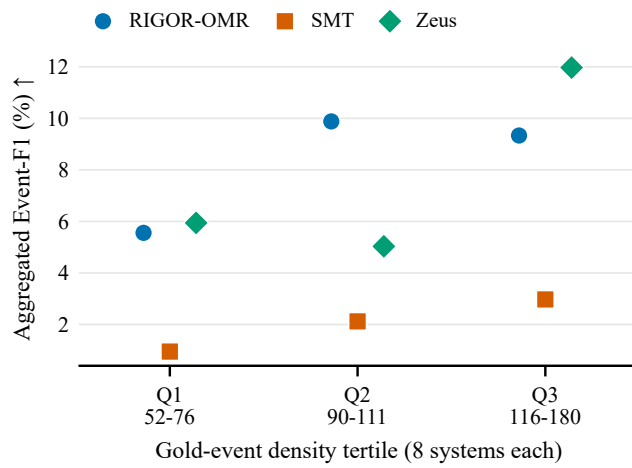


Figure A2: Exploratory Debussy density stratification. Points show aggregated Event-F1 within predeclared gold-event tertiles of eight systems each. No confidence intervals are shown because tertile-level intervals were not estimated; the non-monotonic pattern is descriptive rather than confirmatory.